



MB501P

(PRACTICAL)

Collected Exam Questions

With Comprehensive Answers

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"When we lift others, Allah lifts us."  

Prepared by: **M. Kashif**

Program: BS Zoology

University: Virtual University of Pakistan

 03064963933  burhan.wani888@gmail.com

 WhatsApp Group: <https://chat.whatsapp.com/LOsovr0yLIm6OtYldfW1jT>

Total Questions: 47 | Complete Answers Included



QUESTIONS & ANSWERS



Q1. Distinct features of plant cells.

Answer:

Plant cells have several features that make them different from animal cells:

- 1. Cell wall:** A rigid outer covering made of cellulose that gives the cell its fixed shape and provides support and protection.
- 2. Large central vacuole:** A big fluid-filled sac in the centre that stores water, nutrients, and waste. It keeps the cell firm (turgid).
- 3. Plastids:** Special organelles like chloroplasts (for photosynthesis), chromoplasts (give colour to flowers and fruits), and leucoplasts (store food like starch).
- 4. Chloroplasts:** Contain the green pigment chlorophyll that captures sunlight to make food by photosynthesis.
- 5. Fixed shape:** Because of the cell wall, plant cells are usually rectangular or box-shaped.
- 6. No centrioles:** Most plant cells do not have centrioles (used in cell division in animal cells).
- 7. Plasmodesmata:** Tiny channels in the cell wall that connect neighbouring plant cells and allow materials to pass between them.

Q2. Primary objective of meiosis. How is it different from mitosis?

Answer:

Primary objective of meiosis: The main goal of meiosis is to produce sex cells (gametes like sperm and eggs) that have half the number of chromosomes (haploid). When two gametes join during fertilisation, the full chromosome number (diploid) is restored.

Differences from mitosis:

- 1. Purpose:** Mitosis is for growth and repair. Meiosis is for making sex cells.
- 2. Number of divisions:** Mitosis has one division. Meiosis has two divisions (meiosis I and meiosis II).
- 3. Cells produced:** Mitosis makes 2 identical daughter cells. Meiosis makes 4 genetically different daughter cells.
- 4. Chromosome number:** Mitosis keeps the same chromosome number (diploid). Meiosis cuts it in half (haploid).
- 5. Crossing over:** Happens in meiosis I (prophase I) but not in mitosis.
- 6. Genetic variation:** Mitosis produces identical cells. Meiosis produces genetically unique cells due to crossing over and independent assortment.

Q3. Overview of meiosis II.

Answer:

Meiosis II is the second division in meiosis. It is very similar to mitosis.

Prophase II: Chromosomes condense again. A new spindle forms. The nuclear membrane breaks down.

Metaphase II: Chromosomes line up in a single row along the middle of the cell (equator).

Anaphase II: The sister chromatids are pulled apart to opposite ends of the cell.

Telophase II: Chromatids reach the poles. Nuclear membranes form around each set. The cytoplasm divides (cytokinesis).

Result: Four haploid cells are produced, each with half the number of chromosomes. In males, all four become sperm cells. In females, usually only one becomes a functional egg.

Q4. Overview of meiosis I.**Answer:**

Meiosis I is the first division and is called the **reduction division** because it cuts the chromosome number in half.

Prophase I: Chromosomes condense. Homologous chromosomes pair up (synapsis) forming tetrads. Crossing over happens here, where pieces of DNA are exchanged between homologous chromosomes. This creates genetic variation.

Metaphase I: Homologous pairs line up along the middle of the cell in two rows. Which way each pair faces is random (independent assortment).

Anaphase I: Homologous chromosomes are pulled apart to opposite poles. Unlike mitosis, sister chromatids stay together.

Telophase I: Chromosomes reach the poles. Nuclear membranes may reform. The cell divides into two cells.

Result: Two haploid cells are produced, each with one chromosome from each homologous pair.

Q5. Differentiate between acidic and basic dyes.**Answer:****Acidic dyes:**

- They have a **negative charge** on the coloured part (anion is the chromophore).
- They stain positively charged (basic) structures in the cell, like cytoplasm and some proteins.
- **Examples:** Eosin, Congo red, Acid fuchsin, Nigrosin.
- They are used for background (negative) staining because they are repelled by the negatively charged bacterial surface.

Basic dyes:

- They have a **positive charge** on the coloured part (cation is the chromophore).
- They stain negatively charged (acidic) structures like DNA, RNA, and the bacterial cell surface.
- **Examples:** Crystal violet, Methylene blue, Safranin, Basic fuchsin.
- They are the most commonly used dyes in microbiology because most cell parts carry a negative charge.

Q6. What are different regions of onion root tips?**Answer:**

The onion root tip has four main regions, starting from the very tip going upward:

- 1. Root cap:** The very tip that protects the growing part as it pushes through the soil. Its cells are replaced as they wear off.
- 2. Zone of cell division (Meristematic zone):** Just above the root cap. This is where cells are actively dividing by mitosis. This is the region we study to observe mitosis under the microscope.
- 3. Zone of elongation:** Above the dividing zone. Here the cells stretch and grow longer, pushing the root deeper into the soil.
- 4. Zone of differentiation (Maturation zone):** The top region where cells specialise into different types like root hair cells, xylem, and phloem.

Q7. Define typical structures found in bacteria and their functions.**Answer:**

Bacteria are prokaryotic cells with the following main structures:

- 1. Cell wall:** A rigid layer (made of peptidoglycan) that gives the cell its shape and protects it from bursting.
- 2. Cell membrane:** A thin layer inside the cell wall that controls what enters and leaves the cell.
- 3. Cytoplasm:** The jelly-like substance inside where all chemical reactions take place.
- 4. Nucleoid:** The region where the circular DNA (chromosome) is found. Bacteria have no true nucleus.
- 5. Ribosomes (70S):** Small structures that make proteins.
- 6. Plasmids:** Small extra circles of DNA that carry bonus genes like antibiotic resistance.
- 7. Flagella:** Whip-like tails used for movement.
- 8. Pili/Fimbriae:** Short hair-like projections. Pili help in conjugation (DNA transfer). Fimbriae help the bacterium stick to surfaces.
- 9. Capsule/Slime layer:** A sticky outer coat that protects the bacterium from drying out and from the immune system.

Q8. Name the different types of microscopic regions of root tips in onion.**Answer:**

When you look at an onion root tip under the microscope, you can see these regions:

- 1. Root cap region:** Cells here are irregular and loosely arranged. They protect the tip.
- 2. Meristematic region (zone of division):** Cells are small, tightly packed, have large nuclei, and many are in different stages of mitosis. This is the best area to study cell division.
- 3. Elongation region:** Cells are longer and more rectangular than those in the meristematic region.
- 4. Maturation region:** Cells are fully grown and have specialised shapes. Root hairs may be visible here.

Q9. Coarse adjustment and fine adjustment in detail.

Answer:

These are the two focusing knobs on a microscope:

Coarse adjustment knob:

- It is the **larger knob**, usually on the side of the microscope arm.
- It moves the stage (or body tube) up and down quickly over a large distance.
- It is used first to bring the specimen roughly into focus.
- It should only be used with the **low power** objective lens (4x or 10x). Using it with high power can crack the slide or damage the lens.

Fine adjustment knob:

- It is the **smaller knob**, usually next to or below the coarse knob.
- It moves the stage very slightly and slowly for precise focusing.
- It is used after the coarse knob to get a sharp, clear image.
- It is the only knob that should be used with **high power (40x)** and **oil immersion (100x)** lenses.

Rule: Always start focusing with the coarse knob on low power, then switch to fine adjustment for sharpness.

Q10. Why do we stain microbes? / Why do we need staining for observing microbes?

Answer:

We stain microbes because:

1. Most microbes are colourless (transparent), so they are almost invisible under a light microscope without staining.
2. Staining adds colour to the cells, making them easy to see against the bright background.
3. Different stains can show different structures inside the cell, like the nucleus, cell wall, or flagella.
4. Staining helps us tell different types of bacteria apart. For example, Gram staining separates bacteria into Gram-positive (purple) and Gram-negative (pink).
5. It helps identify the shape (cocci, bacilli, spirilla) and arrangement (chains, clusters, pairs) of bacteria.
6. Special stains can show specific features like capsules, endospores, and acid-fast properties.

Q11. What are fungal cells?

Answer:

Fungal cells are eukaryotic cells that belong to the kingdom Fungi. **Key features include:**

1. **Cell wall:** Made of chitin (not cellulose like plants). Chitin is a tough, flexible material.
2. **No chloroplasts:** Fungi cannot make their own food by photosynthesis. They get food by absorbing nutrients from their surroundings (they are heterotrophs).
3. **Nucleus:** They have a true nucleus with a nuclear membrane, like all eukaryotes.
4. **Vacuoles:** They have vacuoles, but usually smaller than in plant cells.
5. **Cell membrane:** Contains a special lipid called ergosterol (animals have cholesterol instead).
6. **Body forms:** Some fungi are single-celled (like yeast). Others are multicellular with thread-like structures called hyphae, which together form a mass called mycelium.

7. Reproduction: They reproduce by spores (sexual and asexual). **Examples:** yeast, mushrooms, moulds, and Aspergillus.

Q12. Parts of root tip.

Answer:

The root tip has four main parts from bottom to top:

- 1. Root cap:** The protective covering at the very tip. It shields the delicate growing cells and produces a slimy substance that helps the root push through soil.
- 2. Meristematic zone (zone of division):** Just behind the root cap. Cells here divide rapidly by mitosis. They are small, cube-shaped, and have large, dark nuclei.
- 3. Zone of elongation:** Above the meristematic zone. Cells absorb water and stretch to become longer, which pushes the root tip forward.
- 4. Zone of differentiation (maturation):** The uppermost region. Cells here stop growing and take on special roles. Root hair cells grow out from here to absorb water and minerals from the soil.

Q13. Why do we use onion root tips for mitosis?

Answer:

Onion root tips are ideal for studying mitosis because:

- 1. Active cell division:** The meristematic zone at the tip has cells that are constantly dividing, so we can see all stages of mitosis at once.
- 2. Large chromosomes:** Onion cells have relatively large chromosomes ($2n = 16$) that are easy to see under the microscope when stained.
- 3. Easy to prepare:** Onion root tips are simple to grow (just place an onion on water), cut, and stain for a slide.
- 4. No complex tissue:** The root tip cells are simple and not covered by tough outer layers, making them easy to squash into a thin layer for viewing.
- 5. Cheap and available:** Onions are cheap and easy to find all year round.
- 6. Good staining:** They take up stains like acetocarmine or aceto-orcein very well, making chromosomes clearly visible.

Q14. Functions of condenser and diaphragm in detail.

Answer:

Both are found below the stage of the microscope and work together to control light:

Condenser:

- It is a **lens system** located under the stage.
- Its job is to collect and focus the light from the light source onto the specimen.
- By moving the condenser **up**, the light becomes more concentrated (brighter). Moving it **down** spreads the light (dimmer).
- A properly adjusted condenser gives even, bright lighting across the whole field of view, which improves image quality and resolution.

Diaphragm (Iris diaphragm):

- It is a ring of overlapping metal plates located inside or just below the condenser.
- It controls the **amount of light** passing through the specimen by opening or closing the aperture.
- **Opening** the diaphragm lets more light through (brighter image). **Closing** it reduces light (more contrast).
- Adjusting the diaphragm helps get the right balance between brightness and contrast, which is important for seeing fine details.

Q15. What do you know about simple staining?**Answer:**

Simple staining is a basic staining technique that uses only **one dye** to colour the cells.

Purpose: It is used to see the shape, size, and arrangement of microorganisms. It does not separate bacteria into types.

Common dyes used: Methylene blue, Crystal violet, Safranin, Carbol fuchsin. These are all basic dyes (positively charged).

Procedure: (1) Make a thin smear of bacteria on a clean glass slide. (2) Air dry it. (3) Heat fix by passing the slide through a flame. (4) Cover with the dye for 30–60 seconds. (5) Wash gently with water. (6) Blot dry and observe under the microscope.

Advantages: Quick, easy, and cheap. Good for seeing basic cell features.

Limitation: It cannot tell the difference between types of bacteria (all cells stain the same colour).

Q16. Write significance of cell division.**Answer:**

Cell division is extremely important for life. Its significance includes:

- 1. Growth:** Organisms grow from a single cell to a complex body by producing new cells through division.
- 2. Repair and healing:** When tissues are damaged (like a cut on the skin), cell division produces new cells to replace the damaged ones.
- 3. Replacement:** Old, worn-out cells (like skin cells and blood cells) are constantly replaced by new ones through cell division.
- 4. Reproduction:** In single-celled organisms (like bacteria and amoeba), cell division is the method of reproduction. In multicellular organisms, meiosis produces sex cells for sexual reproduction.
- 5. Genetic continuity:** Mitosis ensures that each new cell gets an exact copy of the DNA, keeping the genetic information the same.
- 6. Genetic variation:** Meiosis produces cells with different combinations of genes, which is important for evolution and adaptation.

Q17. Write contrasting features of plant cell wall and cell membrane.**Answer:**

Cell Wall:

- Made of **cellulose** in plants.
- **Rigid and thick.**
- Found **outside** the cell membrane.
- **Fully permeable** (allows all molecules to pass through).
- Provides shape, strength, and protection to the cell.
- Found only in plant cells, fungi, and bacteria (not in animal cells).
- **Non-living** structure.

Cell Membrane:

- Made of a **phospholipid bilayer** with proteins.
- **Thin, flexible, and soft.**
- Found just **inside** the cell wall (in plants) or as the outer boundary (in animal cells).
- **Selectively permeable** (controls what enters and exits the cell).
- Controls transport of materials, cell signalling, and maintains the internal environment.
- Found in **all living cells.**
- **Living** structure.

Q18. What are plastids? Their types and functions.

Answer:

Plastids are special organelles found only in plant cells (and some algae). They are involved in making and storing important substances.

Types and functions:

1. Chloroplasts (green plastids):

- Contain the green pigment chlorophyll.
- Carry out **photosynthesis** – they use sunlight, water, and CO₂ to make glucose (food) and oxygen.
- Found mainly in leaves and green parts of the plant.

2. Chromoplasts (coloured plastids):

- Contain non-green pigments like carotenoids (yellow, orange, red).
- Give colour to flowers, fruits, and autumn leaves.
- Help attract insects and birds for pollination and seed spreading.

3. Leucoplasts (colourless plastids):

- Have no pigment. Found in non-green parts like roots and seeds.
- **Store food:** amyloplasts store starch, elaioplasts store fats/oils, proteinoplasts store proteins.

Note: All plastids can change from one type to another. For example, green tomatoes (chloroplasts) turn red (chromoplasts) as they ripen.

Q19. What is cytokinesis? And why is it essential?

Answer:

Cytokinesis is the division of the cytoplasm that happens after the nucleus has divided (after mitosis or meiosis). It splits one cell into two separate daughter cells.

In animal cells: A cleavage furrow (a pinch) forms around the middle of the cell and tightens until the cell is split into two.

In plant cells: A cell plate forms in the middle of the cell from vesicles. This plate grows outward until it reaches the cell membrane, dividing the cell into two. The cell plate later becomes the new cell wall.

Why it is essential:

1. Without cytokinesis, the cell would have two nuclei but not be properly divided, which would cause problems.
2. It ensures each new cell gets its own set of organelles and cytoplasm.
3. It completes the cell division process and allows the new cells to function independently.
4. It is necessary for growth, repair, and reproduction to work properly.

Q20. Distinct features of protozoans.

Answer:

Protozoans are single-celled eukaryotic organisms. **Their main features are:**

1. **Unicellular:** They are made of just one cell that does all the jobs needed for life.
2. **Eukaryotic:** They have a true nucleus enclosed in a nuclear membrane, plus membrane-bound organelles.
3. **Heterotrophic:** They cannot make their own food. They eat other organisms or absorb nutrients from their environment.
4. **Motile:** Most can move using different structures: **cilia** (e.g., Paramecium), **flagella** (e.g., Trypanosoma), **pseudopodia/false feet** (e.g., Amoeba), or some are non-motile (e.g., Plasmodium in some stages).
5. **Live in moist places:** Found in water, soil, and inside other organisms (some are parasites).
6. **Reproduction:** Asexual (by binary fission) and sometimes sexual (by conjugation in Paramecium).
7. **No cell wall:** Unlike bacteria, plants, and fungi, protozoans do not have a cell wall. They only have a cell membrane.
8. **Examples:** Amoeba, Paramecium, Plasmodium (causes malaria), Trypanosoma (causes sleeping sickness).

Q21. What are leucoplasts? Their types and functions.

Answer:

Leucoplasts are colourless (non-pigmented) plastids found in the non-green parts of plants, like roots, seeds, and underground stems.

Their main job is to store food. There are three types:

1. **Amyloplasts:** Store starch (the most common type). Found in potatoes, rice grains, and roots. They convert glucose into starch for long-term storage.
2. **Elaioplasts (Oleoplasts):** Store fats and oils. Found in seeds and nuts like groundnuts, sunflower seeds, and coconuts.
3. **Proteinoplasts (Aleuroplasts):** Store proteins. Found in seeds of pulses and legumes like lentils and beans.

Leucoplasts can change into other types of plastids when exposed to light. For example, potato leucoplasts can turn into chloroplasts when the potato turns green.

Q22. Differentiate between differential stain and simple stain.**Answer:****Simple stain:**

- Uses only **one dye**.
- Stains all cells the **same colour**.
- Only shows the shape, size, and arrangement of cells.
- **Cannot** tell different types of bacteria apart.
- Quick and easy to do.
- **Examples:** Using methylene blue or crystal violet alone.

Differential stain:

- Uses **two or more dyes**.
- Stains different types of cells in **different colours**.
- Can separate bacteria into groups based on differences in their cell structure (like cell wall).
- More complex but gives much more information.
- **Examples:** Gram staining (separates Gram+ purple from Gram- pink), Acid-fast staining (separates acid-fast red from non-acid-fast blue).

Q23. Components of a stain.**Answer:****A stain (dye) is made up of several components:**

- 1. Chromophore:** The part of the dye molecule that gives it colour. It is the chemical group responsible for absorbing light of a specific wavelength.
- 2. Auxochrome:** The part that helps the dye bind to the cell. It gives the dye its charge (positive or negative) and increases its ability to stick to cell structures.
- 3. Solvent:** The liquid in which the dye is dissolved, usually water (aqueous) or alcohol.
- 4. Mordant (sometimes):** A chemical that helps fix the dye to the cell or intensifies the colour. Example: Iodine is used as a mordant in Gram staining to lock crystal violet inside the cells.
- 5. Decolouriser (in some techniques):** A chemical that removes the dye from certain cells. Example: Alcohol/acetone in Gram staining removes crystal violet from Gram-negative cells.
- 6. Counterstain (in some techniques):** A second dye of a different colour used to stain cells that lost the first dye. Example: Safranin in Gram staining colours the Gram-negative cells pink.

Q24. Sub-phases of interphase.**Answer:**

Interphase is the longest phase of the cell cycle. The cell spends most of its time here, preparing for division. **It has three sub-phases:**

1. G1 phase (Gap 1 / First growth phase):

- The cell grows in size and makes proteins and organelles.
- It carries out its normal functions.

- The cell decides whether to divide or enter a resting state (G₀).

2. S phase (Synthesis phase):

- **DNA replication** happens here – each chromosome is copied to make two identical copies (sister chromatids) joined at the centromere.
- After this phase, the cell has double the amount of DNA.

3. G₂ phase (Gap 2 / Second growth phase):

- The cell continues to grow and makes more proteins, especially those needed for cell division (like tubulin for the spindle).
- It checks for any errors in the copied DNA and repairs them before entering mitosis.

After G₂, the cell enters the **M phase** (mitosis and cytokinesis).

Q25. Differentiate between resolving power and magnification power.

Answer:

Magnification power:

- It is the ability of the microscope to make an object **appear larger** than its actual size.
- **Calculated as:** Total magnification = Eyepiece magnification x Objective magnification. For example: 10x eyepiece x 40x objective = 400x total.
- Increasing magnification alone does not always improve the image – it may just make a blurry image bigger.

Resolving power (Resolution):

- It is the ability of the microscope to show two very close points as **separate and distinct**.
 - Better resolution means sharper, clearer images with more detail.
 - A light microscope has a resolving power of about 0.2 micrometres (200 nm). An electron microscope can resolve down to about 0.1 nm.
- Key difference:** Magnification makes things look **bigger**. Resolution makes things look **clearer**. A good microscope needs both high magnification AND high resolution.

Q26. Characteristic features of prokaryotic and eukaryotic cells.

Answer:

Prokaryotic cells (e.g., bacteria):

- **No true nucleus** – DNA floats freely in the cytoplasm in a region called the nucleoid.
- No membrane-bound organelles (no mitochondria, ER, Golgi, etc.).
- **Smaller** in size (0.1–5 micrometres).
- Have **70S ribosomes** (smaller).
- Cell wall made of **peptidoglycan**.
- DNA is **circular** and naked (no histones).
- Divide by **binary fission**.

Eukaryotic cells (e.g., animals, plants, fungi):

- Have a **true nucleus** enclosed in a nuclear membrane.
- Have **membrane-bound organelles** (mitochondria, ER, Golgi, lysosomes, etc.).
- **Larger** in size (10–100 micrometres).

- Have **80S ribosomes** (larger).
- Cell wall varies: cellulose in plants, chitin in fungi, absent in animals.
- DNA is **linear** and wrapped around histones (chromosomes).
- Divide by **mitosis or meiosis**.

Q27. Compare and contrast cell wall and cell membrane.

Answer:

Similarities:

- Both surround and protect the cell.
- Both are present in plant cells.
- Both play roles in controlling what enters and exits the cell (though in different ways).

Differences:

Cell wall: Rigid, thick, made of cellulose (plants) or chitin (fungi) or peptidoglycan (bacteria). Found outside the cell membrane. Fully permeable. Non-living. Gives shape and support. Not present in animal cells.

Cell membrane: Flexible, thin, made of a phospholipid bilayer with proteins. Found just inside the cell wall or as the outer boundary in animal cells. Selectively permeable. Living. Controls what goes in and out. Present in all cells.

Q28. What is cell wall and cell membrane?

Answer:

Cell wall: It is a strong, rigid outer covering found in plant cells, fungi, and bacteria. In plants, it is made of cellulose. It gives the cell its shape, provides support and protection, and prevents the cell from bursting when it absorbs too much water. It is fully permeable, meaning all molecules can pass through it.

Cell membrane (Plasma membrane): It is a thin, flexible layer made of a phospholipid bilayer with embedded proteins. It is found in all living cells (inside the cell wall in plants, or as the outer layer in animal cells). It is selectively permeable, meaning it controls which substances can enter or leave the cell. It is also involved in cell signalling, transport, and maintaining the cell's internal balance.

Q29. Composition and functions of cell wall in plants.

Answer:

Composition:

- The main component is **cellulose**, a strong polysaccharide made of long chains of glucose molecules.
- Other components include **hemicellulose**, **pectin** (which glues cells together in the middle lamella), and sometimes **lignin** (in woody plants for extra strength).
- Some cell walls also contain proteins and waxes.

Functions:

1. **Provides shape:** The rigid wall gives the cell a fixed, rectangular shape.
2. **Support:** It supports the plant body and keeps the plant upright.
3. **Protection:** It protects the cell from physical damage and pathogens.
4. **Prevents bursting:** When the cell absorbs too much water, the wall resists further expansion and stops the cell from bursting (lysis).
5. **Transport:** Water and small molecules can move freely through the cell wall.
6. **Communication:** Plasmodesmata (channels through the wall) allow neighbouring cells to exchange materials.

Q30. Contrasting features between the cell wall of a plant cell and fungi. Also their composition and functions.

Answer:

Plant cell wall:

- Made of **cellulose** (a polysaccharide of glucose).
- Also contains hemicellulose, pectin, and sometimes lignin.
- Rigid and provides structural support.
- Has **plasmodesmata** for communication between cells.
- **Functions:** Gives shape, provides support, prevents bursting, and allows transport.

Fungal cell wall:

- Made of **chitin** (a polymer of N-acetylglucosamine). This is the same material found in insect exoskeletons.
- Also contains glucans, mannans, and glycoproteins.
- Flexible yet tough.
- Does not have plasmodesmata. Fungal cells communicate through **pores in their septa** (cross-walls in hyphae).
- **Functions:** Provides shape and protection, resists environmental stress, and helps the fungus absorb nutrients.

Key difference: Plant walls use **cellulose**; fungal walls use **chitin**. This is one reason antifungal drugs can target chitin without harming plant or human cells.

Q31. What is cyclosis?

Answer:

Cyclosis (also called **cytoplasmic streaming**) is the continuous, circular movement of the cytoplasm inside a living cell.

The cytoplasm flows around the cell in a stream-like pattern, carrying organelles, nutrients, and other molecules with it.

It is powered by the interaction between the protein **actin** (microfilaments) and the motor protein **myosin**, using energy from ATP.

Functions: (1) It helps move materials (like nutrients and organelles) around the cell, especially in large cells where diffusion alone would be too slow. (2) It helps distribute chloroplasts to areas where they can get the most light for photosynthesis.

Cyclosis can be easily observed in cells of aquatic plants like **Elodea** (water weed) or the stamen hair cells of **Tradescantia**.

Q32. Which flower is used to study cyclosis and why?

Answer:

The flower of **Tradescantia** (also known as spiderwort) is commonly used to study cyclosis.

Specifically, the stamen hair cells are used because:

1. The cells are long, thin, and arranged in a single row (chain), making them easy to see under the microscope.
2. The cytoplasm in these cells is actively streaming, so the movement is clearly visible.
3. The cells are transparent, allowing you to see the movement of organelles and chloroplasts without staining.
4. The cells are large enough to observe the circular flow pattern of cytoplasm.

Alternatively, Elodea (Hydrilla) leaf cells are also commonly used for cyclosis observation, where you can see chloroplasts moving along the cell edges in a streaming pattern.

Q33. Differentiate between mitosis and meiosis on the basis of their division and how many cells they produce.

Answer:

Mitosis:

- **Type of division:** One nuclear division (equational division).
- **Cells produced:** 2 daughter cells.
- **Chromosome number:** Same as the parent cell (diploid → diploid). Example: $2n \rightarrow 2n$.
- **Genetic makeup:** Daughter cells are genetically identical to the parent cell.
- **Where:** In body (somatic) cells for growth and repair.

Meiosis:

- **Type of division:** Two nuclear divisions (meiosis I = reduction, meiosis II = equational).
- **Cells produced:** 4 daughter cells.
- **Chromosome number:** Half of the parent cell (diploid → haploid). Example: $2n \rightarrow n$.
- **Genetic makeup:** Daughter cells are genetically different from each other and from the parent cell.
- **Where:** In reproductive organs to produce sex cells (gametes – sperm and eggs).

Q34. Name the reagents that are used in Gram staining.

Answer:

Gram staining uses four reagents in this order:

1. **Crystal violet (Primary stain):** A purple/violet basic dye that stains all bacterial cells purple.
2. **Gram's iodine (Mordant):** It forms a complex with crystal violet inside the cells, locking the dye in. This makes the staining stronger.

3. Alcohol/Acetone (Decolouriser): It washes the crystal violet-iodine complex out of Gram-negative cells (because their thin cell wall cannot hold it). Gram-positive cells keep the purple colour because their thick peptidoglycan wall traps the dye.

4. Safranin (Counterstain): A pink/red basic dye that stains the now colourless Gram-negative cells pink. Gram-positive cells remain purple because they already hold crystal violet.

Result: Gram-positive = purple/violet. Gram-negative = pink/red.

Q35. Different microscopes used in the lab and their functions.

Answer:

1. Light/Compound microscope: Uses visible light and glass lenses. Magnifies up to 1000x–1500x. Used for viewing cells, bacteria (after staining), and tissue sections.

2. Stereo/Dissecting microscope: Has lower magnification (10x–45x) and gives a 3D view. Used for viewing larger specimens like insects, plant parts, and dissections.

3. Phase-contrast microscope: Uses special optics to show transparent, unstained living cells by converting differences in thickness and density into visible light and dark areas.

4. Fluorescence microscope: Uses UV light to excite fluorescent dyes or proteins in the specimen. The specimen glows against a dark background. Used in immunology and molecular biology.

5. Electron microscope: Uses beams of electrons instead of light. Very high magnification (up to 2,000,000x) and resolution. **TEM** (Transmission) shows internal cell structure. **SEM** (Scanning) shows 3D surface detail.

6. Dark-field microscope: Uses a special condenser so only light scattered by the specimen reaches the eye. Useful for viewing living, unstained microbes.

Q36. Prophase stage of mitosis. / Describe the events that occur during prophase.

Answer:

Prophase is the first and longest stage of mitosis. **The following events occur:**

1. Chromatin condenses: The long, thin chromatin fibres coil up tightly and become visible as thick chromosomes under the microscope.

2. Chromosomes become visible: Each chromosome can be seen as two sister chromatids joined at the centromere (because DNA was already copied in S phase).

3. Centrioles move apart: In animal cells, the two pairs of centrioles move to opposite ends (poles) of the cell.

4. Spindle fibres form: Protein fibres (microtubules) begin to form between the centrioles, creating the mitotic spindle.

5. Nucleolus disappears: The nucleolus (where ribosomal RNA is made) breaks down and is no longer visible.

6. Nuclear membrane breaks down: The nuclear envelope starts to break apart at the end of prophase, allowing the spindle fibres to attach to the chromosomes.

Q37. Differentiate between plant and animal cells.

Answer:**Plant cells:**

- Have a **cell wall** made of cellulose.
- Have a **large central vacuole** for water storage.
- Have **chloroplasts** for photosynthesis.
- Usually have a fixed, **rectangular shape**.
- Store food as **starch**.
- **No centrioles** (in most).
- Cytokinesis happens by **cell plate formation**.

Animal cells:

- **No cell wall** (only a cell membrane).
- Have **small vacuoles** or none at all.
- **No chloroplasts**.
- Usually **round or irregular** in shape.
- Store food as **glycogen**.
- Have **centrioles** (used in cell division).
- Cytokinesis happens by **cleavage furrow** (pinching in).

Both have: cell membrane, nucleus, cytoplasm, mitochondria, ribosomes, ER, and Golgi body.

Q38. Similarities and dissimilarities between plant and animal cells.**Answer:****Similarities:**

- Both are **eukaryotic** (have a true nucleus with a nuclear membrane).
- Both have a cell membrane, cytoplasm, ribosomes, mitochondria, ER (rough and smooth), Golgi apparatus, and lysosomes.
- Both contain **DNA** as genetic material in chromosomes.
- Both carry out **cellular respiration** in mitochondria.
- Both divide by mitosis and meiosis.

Dissimilarities:

- **Cell wall:** Present in plants (cellulose), absent in animals.
- **Chloroplasts:** Present in plants (for photosynthesis), absent in animals.
- **Vacuole:** Large central vacuole in plants; small or absent in animals.
- **Shape:** Plants are usually rectangular; animals are round or irregular.
- **Centrioles:** Present in animals; absent in most plants.
- **Food storage:** Starch in plants; glycogen in animals.
- **Cytokinesis:** By cell plate in plants; by cleavage furrow in animals.

Q39. Define meiosis. How many chromosomes are produced at the end? What is the number of chromosomes?**Answer:**

Meiosis is a type of cell division that produces sex cells (gametes) with **half** the number of chromosomes as the parent cell.

It involves two rounds of division: **meiosis I** (reduction division) and **meiosis II** (equational division).

Result: One diploid cell ($2n$) produces **four haploid cells (n)**. Each daughter cell has half the chromosome number of the parent.

Example in humans: Human body cells have 46 chromosomes ($2n = 46$). After meiosis, each gamete (sperm or egg) has **23 chromosomes** ($n = 23$). When a sperm (23) fertilises an egg (23), the full number (46) is restored in the baby.

Meiosis also creates **genetic diversity** through crossing over (in prophase I) and independent assortment (in metaphase I).

Q40. Name 3 different bacteria on the basis of their cell wall composition.

Answer:

Based on cell wall composition, bacteria can be grouped as:

1. Gram-positive bacteria: Have a **thick peptidoglycan layer** (20–80 nm) in their cell wall. They also have teichoic acids. They stain **purple/violet** in Gram staining. **Examples:** *Staphylococcus aureus*, *Bacillus subtilis*.

2. Gram-negative bacteria: Have a **thin peptidoglycan layer** (5–10 nm) and an extra **outer membrane** containing lipopolysaccharides (LPS). They stain **pink/red** in Gram staining.

Examples: *Escherichia coli*, *Salmonella typhi*.

3. Acid-fast bacteria (Mycobacteria): Have a unique cell wall rich in **mycolic acids** (waxy lipids) over a thin peptidoglycan layer. This waxy wall makes them resistant to Gram staining and many chemicals. They are identified by the **acid-fast (Ziehl-Neelsen) stain**. **Example:** *Mycobacterium tuberculosis*.

Q41. Write down the features of plant cells.

Answer:

(Same as Q1 – summary)

Plant cells have these key features:

- 1. Cell wall** made of cellulose – gives rigid shape and support.
- 2. Large central vacuole** – stores water, keeps the cell firm.
- 3. Chloroplasts** – carry out photosynthesis using chlorophyll.
- 4. Chromoplasts** – give colour to flowers and fruits.
- 5. Leucoplasts** – store food (starch, oils, proteins).
- 6. Plasmodesmata** – tiny channels connecting plant cells.
- 7. No centrioles** in most plant cells.
- 8. Fixed rectangular shape.**
- 9. Store energy as starch granules.**
- 10. Cytokinesis by cell plate formation.**

Q42. Write the procedure to prepare a slide for human epidermal cells. What cellular components are observed?

Answer:

Procedure:

1. Take a clean glass slide and place a drop of methylene blue stain in the centre.
2. Using a clean, flat toothpick or wooden spatula, gently scrape the inside of your cheek.
3. Spread the scraping onto the slide in the drop of stain. Mix gently to make a thin, even smear.
4. Place a cover slip carefully over the smear to avoid air bubbles.
5. Leave the stain for 1–2 minutes so the cells take up the colour.
6. Blot any excess stain from the edges with a paper towel.
7. Observe under the microscope, starting with low power (10x) and then switching to higher power (40x).

Cellular components observed:

- **Cell membrane:** The outer boundary of the cell (thin outline).
- **Nucleus:** A dark-stained, round structure in the centre of the cell.
- **Cytoplasm:** The stained area inside the cell around the nucleus.
- **Note:** Animal cells (cheek cells) do not have a cell wall, chloroplasts, or a large central vacuole.

Q43. Write the characteristic features of eukaryotic cells.

Answer:

Eukaryotic cells have the following features:

1. **True nucleus:** Their DNA is enclosed inside a double-layered nuclear membrane.
2. **Membrane-bound organelles:** They have organelles like mitochondria, endoplasmic reticulum (ER), Golgi body, lysosomes, and (in plants) chloroplasts.
3. **Linear DNA:** Their DNA is arranged in linear chromosomes wrapped around histone proteins.
4. **Larger size:** They are usually 10–100 micrometres in diameter.
5. **80S ribosomes:** Their ribosomes are larger than those in prokaryotes.
6. **Cytoskeleton:** They have a network of protein filaments (microtubules, microfilaments) that give the cell shape and help with movement.
7. **Cell division:** They divide by mitosis (for body cells) or meiosis (for sex cells).
8. **Examples:** Animal cells, plant cells, fungal cells, and protozoans.

Q44. Differentiate between RER and SER.

Answer:

RER (Rough Endoplasmic Reticulum):

- Has **ribosomes** attached to its surface (looks rough under electron microscope).
- Mainly involved in **making proteins**, especially those that will be exported out of the cell or sent to the cell membrane.
- Well-developed in cells that make a lot of proteins, like antibody-producing cells and pancreatic cells.

- Connected to the nuclear envelope.

SER (Smooth Endoplasmic Reticulum):

- Has **no ribosomes** on its surface (looks smooth).
- Involved in making **lipids (fats)**, **steroid hormones**, and **detoxifying** harmful substances.
- Stores **calcium ions** (important in muscle cells).
- Well-developed in cells that make lipids or detoxify drugs, like liver cells and cells that produce steroid hormones.

Both are connected to each other and are part of the same membrane network inside the cell.

Q45. What is the mordant in Gram staining? And why is it used?

Answer:

The mordant in Gram staining is **Gram's iodine** (an iodine-potassium iodide solution).

What it does: Iodine enters the bacterial cells and forms a large complex with crystal violet (called the **CV-I complex**). This complex is much bigger and harder to wash out than crystal violet alone.

Why it is used:

1. It makes the staining stronger by locking the dye inside the cell.
2. In **Gram-positive cells** (which have a thick peptidoglycan wall), the CV-I complex gets trapped inside even when the decolouriser is added. This is why they stay purple.
3. In **Gram-negative cells** (which have a thin peptidoglycan layer and an outer membrane), the decolouriser dissolves the outer membrane and washes the CV-I complex out. This is why they lose their purple colour and take up the counterstain (safranin) to appear pink.

Without the mordant, the crystal violet would wash out of all cells too easily, and the staining would not work properly.

Q46. What is the usual size of animal cells? Write about the largest and smallest animal cells.

Answer:

Usual size: Most animal cells range from **10 to 30 micrometres** in diameter. This is much bigger than bacteria (1–5 micrometres) but smaller than many plant cells.

Largest animal cell: The **ostrich egg** (ovum) is the largest single animal cell. It can be about 15–18 cm in diameter (including the yolk). In the human body, the largest cell is the **female egg cell (ovum)**, about 100–120 micrometres.

Smallest animal cell: The smallest known animal cells include **sperm cells**. Human sperm cells are about 5 micrometres in length (head only). Among nucleated cells, the smallest are certain **white blood cells (lymphocytes)** at about 7–8 micrometres.

The size of a cell depends on its function. **Nerve cells** can be very long (up to 1 metre) even though their cell body is small. **Red blood cells** are about 7–8 micrometres and have no nucleus.

Q47. Compare and contrast the cell wall of plant cells and fungal cells, highlighting their functions and composition.

Answer:

(Same as Q30 – detailed comparison)

Plant cell wall:

- **Composition:** Mainly cellulose, with hemicellulose, pectin, and sometimes lignin.
- **Structure:** Rigid, layered (primary and secondary wall). Has plasmodesmata for cell-to-cell communication.
- **Functions:** Gives shape, provides strength and support, prevents the cell from bursting, allows water and small molecules to pass through freely.

Fungal cell wall:

- **Composition:** Mainly chitin, with glucans, mannans, and glycoproteins.
- **Structure:** Flexible but strong. Has pores in septa (cross-walls) for communication in multicellular fungi.
- **Functions:** Gives shape and protection, resists drying out and environmental stress, helps in nutrient absorption.

Similarities: Both give the cell its shape and protection. Both are found outside the cell membrane. Both are permeable to water and small molecules.

Key difference: The main building material is **cellulose in plants** and **chitin in fungi**. This structural difference is important in medicine because drugs can target chitin without harming human or plant cells.

✧ **End of Document** ✧

May Allah bless your efforts and make your path easy.

When we lift others, Allah lifts us. 🤝